

Introduction

The purpose of this project is to explore and analyze a polling dataset using data analysis and visualization techniques in Python. The dataset contains information related to election polls, including polling organizations, polling dates, sample sizes, population types, and polling percentages for different candidates. The main objective of this analysis is to understand polling trends better, compare candidate performance, and identify meaningful patterns in the dataset.

To complete this project, we used Google Colab and Python libraries such as pandas, matplotlib, and seaborn. These tools allowed us to organize, explore, and visualize the data efficiently.

The project is divided into three main tasks. The first task focuses on creating a data dictionary and identifying the most important variables in the dataset. The second task involves creating univariate and bivariate visualizations to better understand the data and explore relationships between variables. Finally, the third task focuses on producing more refined and definitive visualizations using appropriate visualization principles such as labeling, annotations, legends, and presentation techniques.

Task 1: Data Dictionary and Important Variables

For the first task, we created a data dictionary to understand the dataset better and organize the information before starting the analysis. This step helped us identify the variables that were most useful for describing the polls, comparing Clinton and Trump, and preparing the exploratory visualizations.

We started by exploring the structure of the dataset to understand the different columns available, the type of information stored in each column, and how the data was organized. We identified both categorical variables, such as the state and population type, and numerical variables, such as polling percentages and sample sizes.

After reviewing the dataset, we selected the variables that we believed were the most important for understanding polling trends and candidate performance. These variables included information about: the location of the poll, the timeline of the poll, the organization conducting the poll, the quality and reliability of the poll, the number of participants surveyed, and the polling percentages for the main candidates.

The main variables selected for the data dictionary included: ***state, startdate, enddate, enddate, pollster, grade, samplesize, population, rawpoll_clinton, rawpoll_trump, adjpoll_clinton, adjpoll_trump.***

The table below summarizes the main variables selected for the analysis.

	Variable	Description	Justification	Data Type
0	state	State where the poll was conducted	Helps identify geographic voting trends	object
1	startdate	Date when the poll started	Helps place the beginning of each poll in the campaign timeline	object
2	enddate	Date when the poll ended	Helps place each poll in the campaign timeline	object
3	pollster	Polling organization	Different pollsters may use different methods	object
4	grade	Poll quality grade	Gives context about the pollster's overall quality rating	object
5	samplesize	Number of participants surveyed	Larger samples generally provide more stable estimates, although sample size alone does not determine poll quality	float64
6	rawpoll_clinton	Raw support percentage for Clinton	Used to analyze Clinton's adjusted polling estimate	float64
7	rawpoll_trump	Raw support percentage for Trump	Used to analyze Trump's adjusted polling estimate	float64
8	adjpoll_clinton	Adjusted support percentage for Clinton	Provides corrected estimates for Clinton	float64
9	adjpoll_trump	Adjusted support percentage for Trump	Provides corrected estimates for Trump	float64
10	population	Type of population surveyed	Helps understand which group was surveyed (adults, registered voters, likely voters, etc.)	object
11	raw_margin_clinton_trump	Difference between Clinton's raw polling percentage and Trump's raw polling percentage	Helps compare the two main candidates directly using the original poll results	float64
12	adj_margin_clinton_trump	Difference between Clinton's adjusted polling percentage and Trump's adjusted polling percentage	Central variable for analyzing who is leading after polling adjustments	float64

These variables were chosen because they give useful insight into how the polls were conducted and how support for the candidates changed across different polls and locations.

To make the comparison between Clinton and Trump easier, we also created two additional variables. The first variable, *raw_margin_clinton_trump*, measures the difference between Clinton's and Trump's original polling percentages before adjustments were applied. The second variable, *adj_margin_clinton_trump*, measures the difference between their adjusted polling percentages after corrections and weighting methods were taken into account. These new variables helped simplify the comparison between the two candidates and made it easier to identify which candidate was leading in a particular poll.

For each variable included in the data dictionary, we provided the variable name, data type, description, and a short explanation of why the variable was important for the analysis. Variables containing descriptive information, such as state, population, and pollster, were classified as categorical variables, while variables representing measurable values, such as polling percentages and sample sizes, were classified as numerical variables.

Overall, Task 1 provided a strong foundation for the rest of the project by helping us organize the dataset, understand the variables, and prepare for the visualization and exploratory analysis stages.

Task 2: Exploratory Visualizations

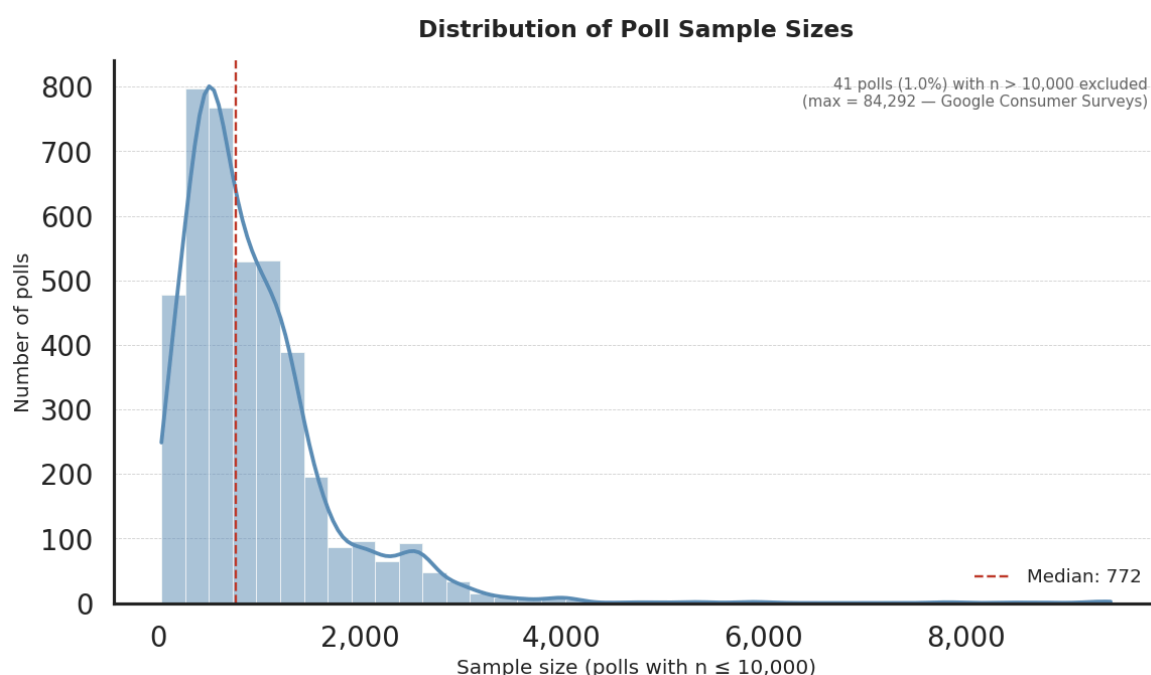
Before moving to the final storytelling visuals, we first used exploratory visualizations to better understand the dataset. At this stage, our goal was not to create the most polished or

definitive charts, but to ask useful questions about the data and identify patterns that could guide the rest of the analysis.

These exploratory graphs helped us examine the structure of the polling data from different angles: sample size, time, polling adjustments, geography, population groups, and state-level adjustment patterns. Some of the charts are more detailed than others, but they are mainly used here as analytical tools. Their purpose is to help us decide which findings are worth developing later into clearer and more refined presentation visuals.

For this reason, the visualizations in this section should be understood as working graphs. They help us explore what is happening in the data before selecting the clearest findings for the final part of the project.

Exploratory Visualization 1: Distribution of Poll Sample Sizes

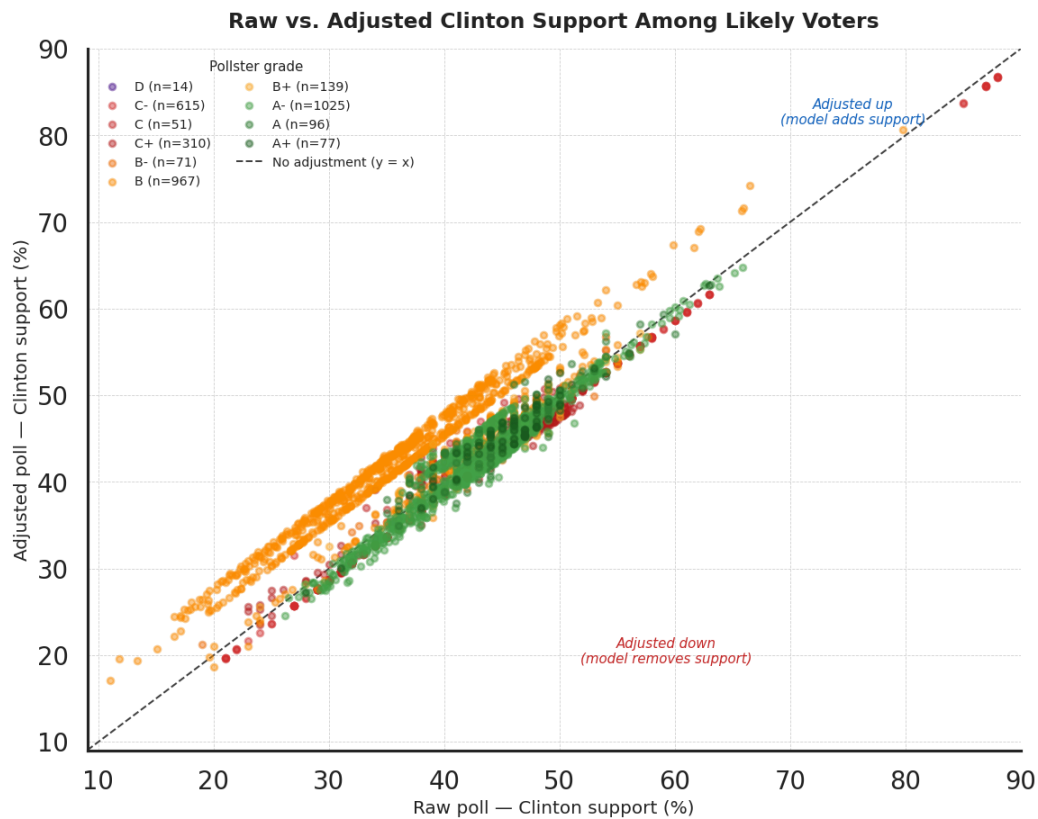


The first visualization explores the distribution of poll sample sizes. We used a histogram because the sample size is a numerical variable, and this type of chart makes it easier to see where most observations are concentrated.

The graph shows that most polls had relatively moderate sample sizes, while a small number of polls had much larger samples. These very large polls stretch the x-axis and make the main distribution harder to read. To make the pattern clearer, we excluded polls above 10,000 respondents from the display, while still documenting this choice directly on the graph. This allowed us to reduce visual clutter without hiding the existence of outliers.

We also added a median line to give a quick reference point for the typical poll size. This is useful because the distribution is right-skewed, meaning that a few very large polls pull the upper range away from most observations. As an exploratory graph, this visualization helps us understand the methodological structure of the dataset before comparing polling results.

Exploratory Visualization 2: Raw vs. Adjusted Clinton Support Among Likely Voters



This visualization compares Clinton's raw polling support with her adjusted polling support among likely voters.

Each point represents one poll, and the colors show the pollster grade. The dashed diagonal line represents the point where the raw and adjusted values are equal. Most points are close to this line, which suggests that Clinton's adjusted polling values were usually similar to her raw polling values. However, some points are above or below the line, meaning that the adjustment process sometimes increased or decreased her estimated support.

This graph is useful because our later analysis relies mostly on adjusted polling variables, so it helps us understand how different those adjusted values are from the original raw poll results.

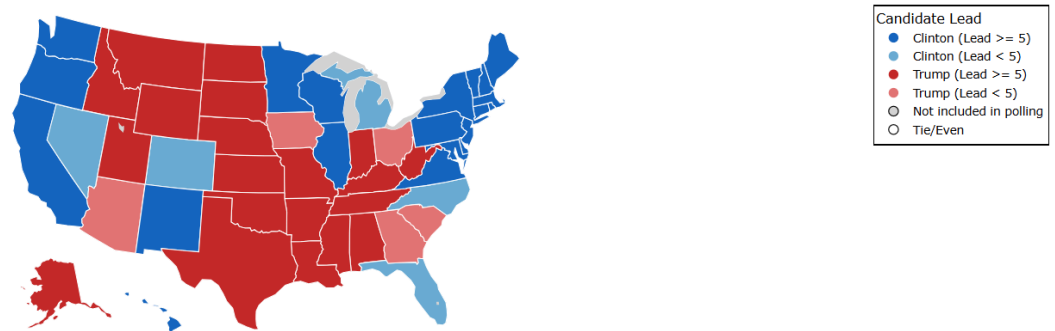
Exploratory Visualization 3: 2016 US elections Raw polling by state

This visualization is a map that shows Trump and Clinton's raw polling support among likely voters. The color of each state represents the color associated with the candidate in the lead. We used the numbers obtained by taking the median of raw polling for each state, as it is more robust against outliers. The intensity of the colours shows whether the gap between the lead candidate and the second place was higher than or equal to 5 or lower.

This visualization is important as it gives us a picture of which candidate is in the lead in which state. It also provides information on how close the gap is, which helps us identify where each candidate still has the potential to lead or to shift election results. As we can

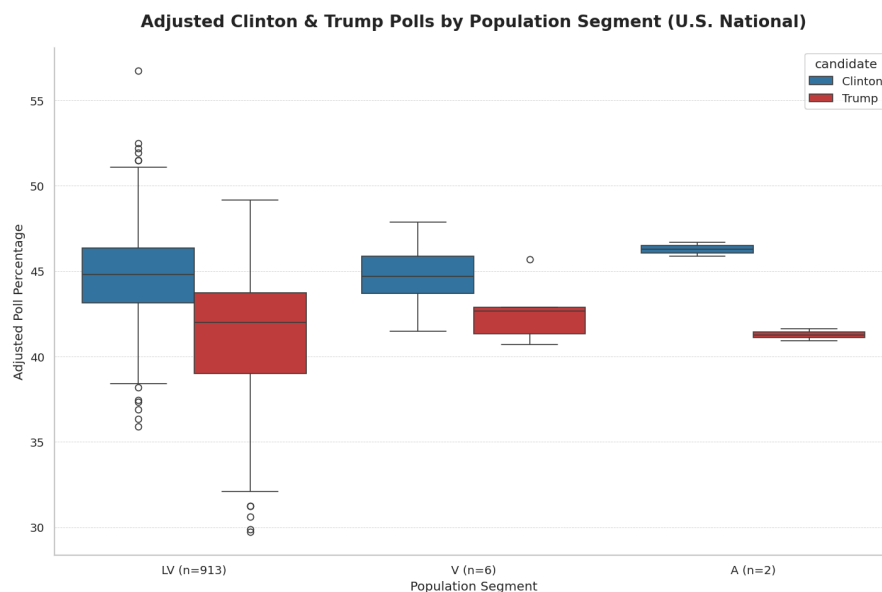
see, in Texas and Oklahoma, Trump is in the lead in terms of support from likely voters, whereas in Maine and New York, Clinton is in the lead. However, for states such as Florida and Georgia, the gap is smaller than 5 votes, which indicates the results could reveal either one as the lead candidate in the elections.

2016 US Election Raw Polling by State (Likely Voters Only)



Link to interactive map (please download to open): [Viz4_Map_Clinton-Trump_raw_polls.html](#)

Exploratory Visualization 4: Adjusted Clinton and Trump Polls by population segment

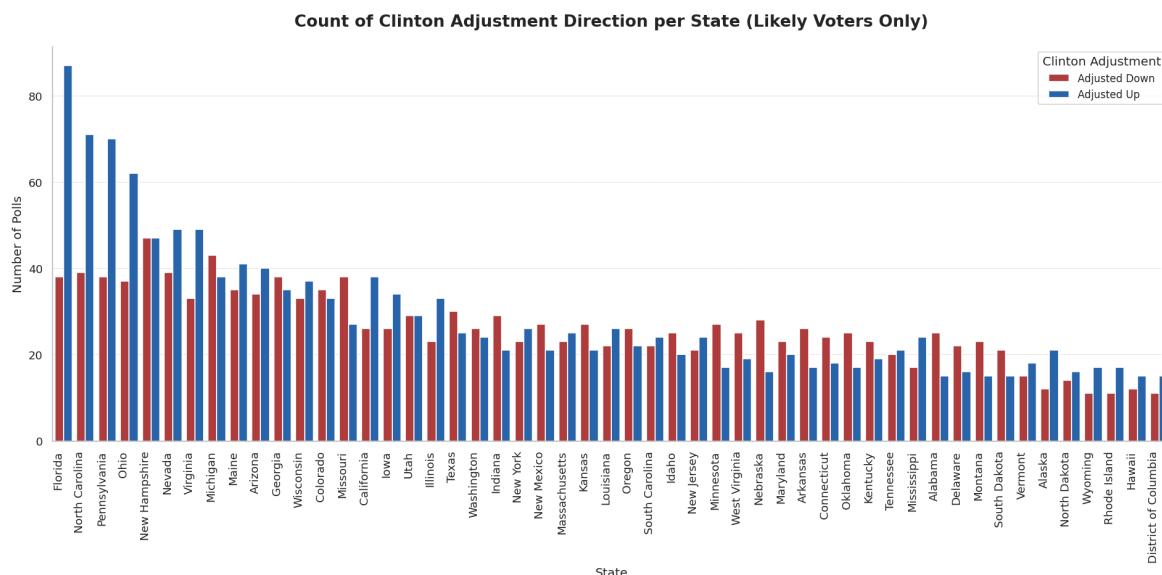


This visualization explores the distribution of adjusted polling for Clinton and Trump for each population segment. We chose a boxplot, because this choice of visualisation not only shows the distribution, but it also helps us identify if there are any outliers in adjusted polling.

The size of the boxes is representative of the overall number of pollsters polling people from the specified population segment. The line in the middle tells us the median adjusted poll for the specific candidate among people from the chosen population segment.

As such, we can see that among likely voters, with all states combined, the median adjusted polling support is higher for Clinton than it is for Trump. With this visual, we see the distribution among likely voters supporting Clinton shows low and high outliers, meaning higher numbers and lower numbers than the norm (here, within the norms are the values within the interquartile range). As for Trump supporters among likely voters, we only see low outliers, meaning some pollsters' adjusted polling results for Trump supporters were lower than the norm.

Exploratory Visualization 5: Clinton and Trump adjustment directions



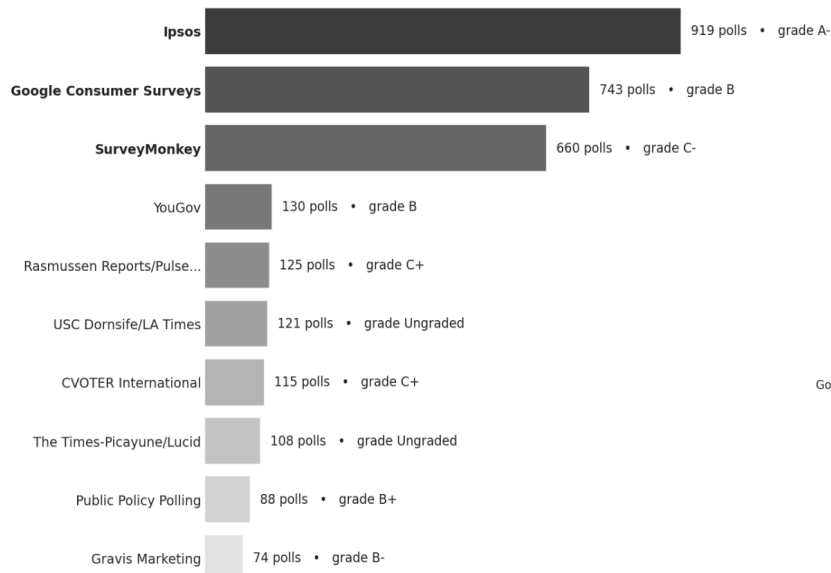
This visualization explores the effect of the adjustments on the polling results per state. It looks at the difference between the raw polling results and the adjusted polling results to identify whether the adjustment lowered or raised the polling results for each state. In blue are the number of adjustments that were higher than the raw polling result for each state, and in red the opposite.

This type of visualization was chosen because we wanted to know how many times the adjustments raised or lowered the results from the raw polling results for each state. For states where the raw polling is frequently lower, this tells us the number of estimated likely voters supporting the specified candidate might be lower than anticipated. On the other hand, results that are frequently adjusted higher than the raw polling might indicate an underestimation of the support towards a candidate among likely voters.

Task 3: Final Storytelling Visualizations

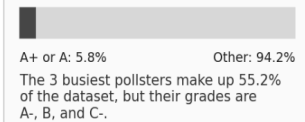
First Visualization :

Who produced most of the polling evidence?

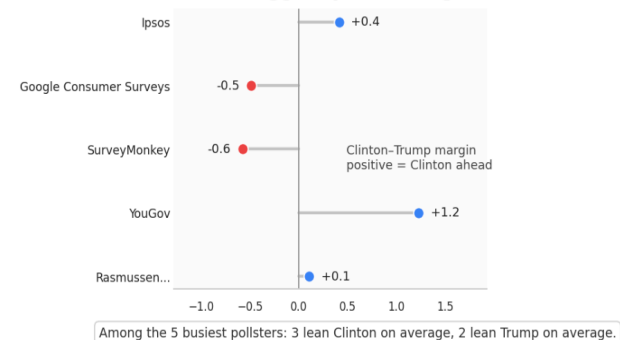


Volume ≠ top grade

Only 5.8% of all poll rows come from A+ or A pollsters.



Do the biggest pollsters agree?



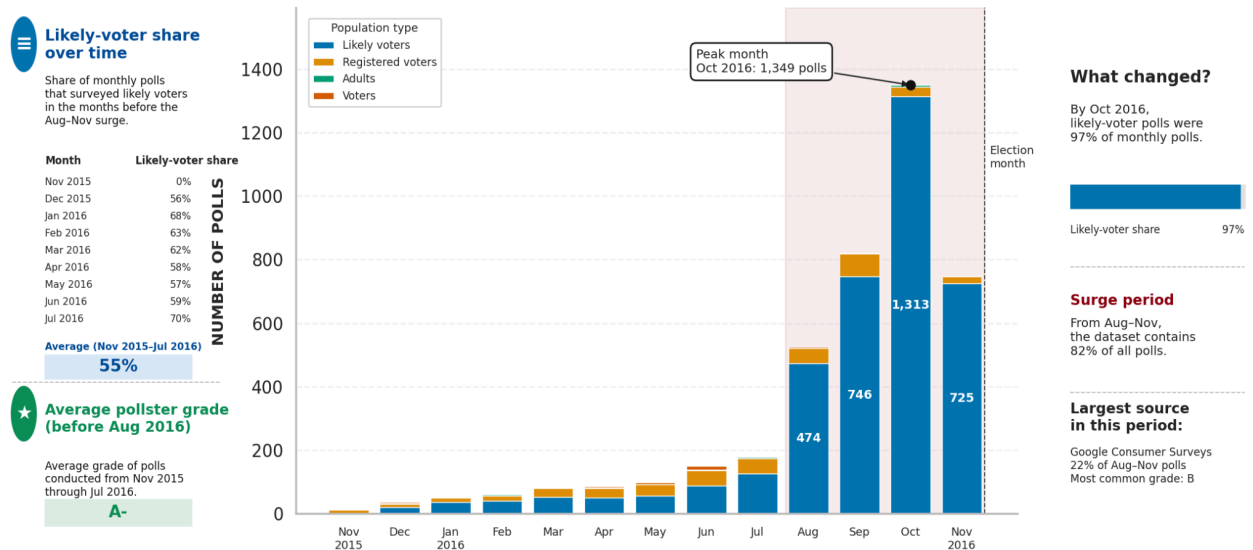
Before interpreting the Clinton–Trump polling results, we first looked at where most of the polling data comes from. This is important because pollsters with more observations have a larger influence on the dataset. As shown in the graph, most polls come from a small number of pollsters, especially Ipsos, Google Consumer Surveys, and SurveyMonkey.

However, high volume does not necessarily mean high quality. The three busiest pollsters are graded A-, B, and C-, while only about 5.8% of all poll rows come from A+ or A pollsters. This suggests that we should be cautious when interpreting the dataset and not blindly treat all polls as equally reliable.

The bottom-right graph shows that the biggest pollsters disagree on the average Clinton–Trump margin. Since they make up a large part of the dataset, this disagreement can shift the overall polling story depending on which pollster dominates the data.

Second Visualization :

Types of voters over time

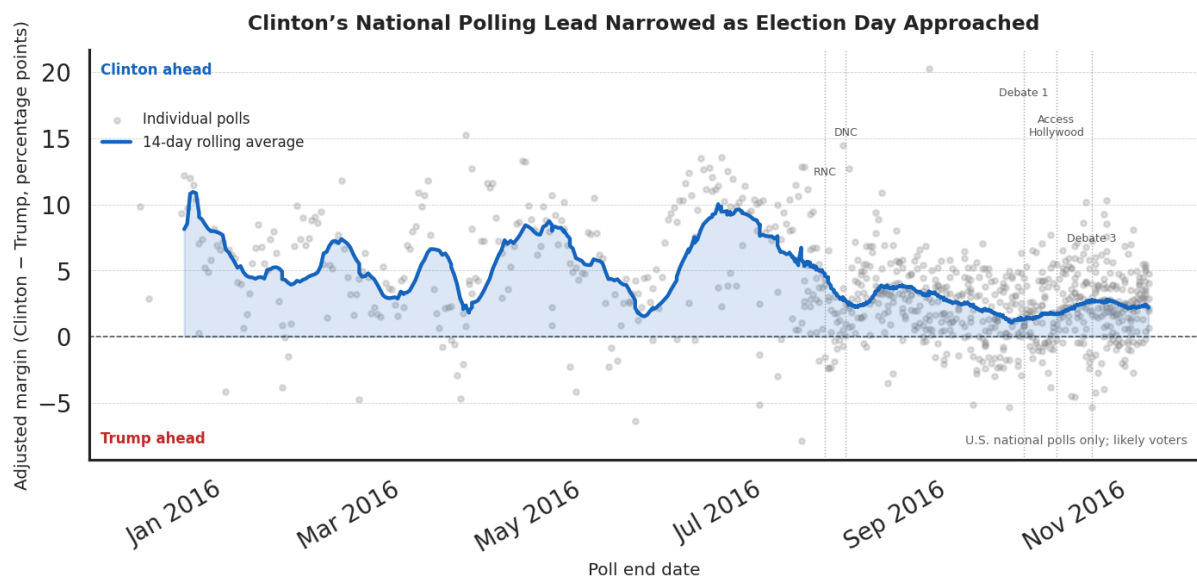


Next, it is important to understand how the composition of surveyed voters changed over time. While likely voters were already the largest group early in the election cycle, they became much more dominant as election day approached. This likely happened because pollsters wanted to focus more on people who were actually expected to vote, especially once the election entered its final months and prediction accuracy became more important.

The graph also shows a major surge in polling activity around August to November 2016, with October becoming the peak polling month in the dataset. This is probably due to rising media attention, increased public interest, and the fact that the election was becoming more competitive and important as voting day got closer. Since this late-election period contains a very large share of all polls, it strongly influences the overall dataset.

The side information also shows that the average pollster grades before August 2016 were still around the A- to B range, meaning the earlier polls were not necessarily low quality. However, the final months were dominated by a smaller number of major polling organizations, meaning that the overall polling narrative near election day may have been shaped more heavily by a few large pollsters than by the dataset as a whole.

Visualization 3: Clinton's National Polling Lead Narrowed as Election Day Approached



Before concluding the polling results, we also looked at how the Clinton–Trump margin changed over time. This is important because the meaning of a poll depends strongly on when it was conducted. A poll from early 2016 does not reflect the same campaign context as a poll taken close to election day.

The graph shows that Clinton was ahead in most national likely-voter polls, since the adjusted margin usually remains above zero. However, her lead was not stable. It was stronger earlier in the campaign, especially around late spring and early summer, but became narrower as election day approached. The 14-day rolling average helps show this pattern more clearly by reducing the noise from individual polls.

This suggests that the main story is not only that Clinton was leading, but that her lead weakened over time. Because the dataset contains many polls from the final months of the campaign, this narrowing margin is important for understanding the overall polling picture.

Conclusion

Overall, this project helped us move from a raw polling dataset to a clearer understanding of the 2016 U.S. election polling story. By first organizing the main variables, then exploring the data visually, we were able to see that the results were shaped by more than just candidate percentages. Pollster volume, sample size, population type, timing, and polling adjustments all played an important role. The final visualizations showed that much of the polling evidence came from a small number of pollsters, that likely-voter polls became especially dominant near Election Day, and that Clinton's national lead narrowed over time. In the end, this project showed us that good visualizations not only make data look clearer; they also help us understand the context behind the numbers.